

Project Background – Geotechnical Summary

Tangier Island is located within the northern portion of the Chesapeake Bay. The island itself is made up of Quaternary alluvial, intertidal, and eolian deposits. Alluvium consists of organic material, clay, sand, and subrounded pebbles and cobbles. Intertidal deposits are composed of organic-rich clays and silts, while beach/eolian deposits consist of poorly sorted fine to medium quartz sand. Tangier Island's formation was heavily influenced by the fluvial system of the Susquehanna River, where a ridge was created along its edge during the last sea-level low-stand around 18,000 years ago. Today, the island's highest point is approximately 6 feet above sea level.

Sediment composition within the federal navigation channel varies by location (Figure 1). Coarser-grained sediments are found along the western side of the channel, beyond the jetty, likely due to fluvial input from nearby rivers. Moving east across the channel, sediments become much finer (less than 50% sand), likely due to lower fluvial input and a lower-energy depositional environment. The 2024 vibrocore data represents subsurface conditions at the time of sampling and are subject to change due to shoal movement and dredging activities.

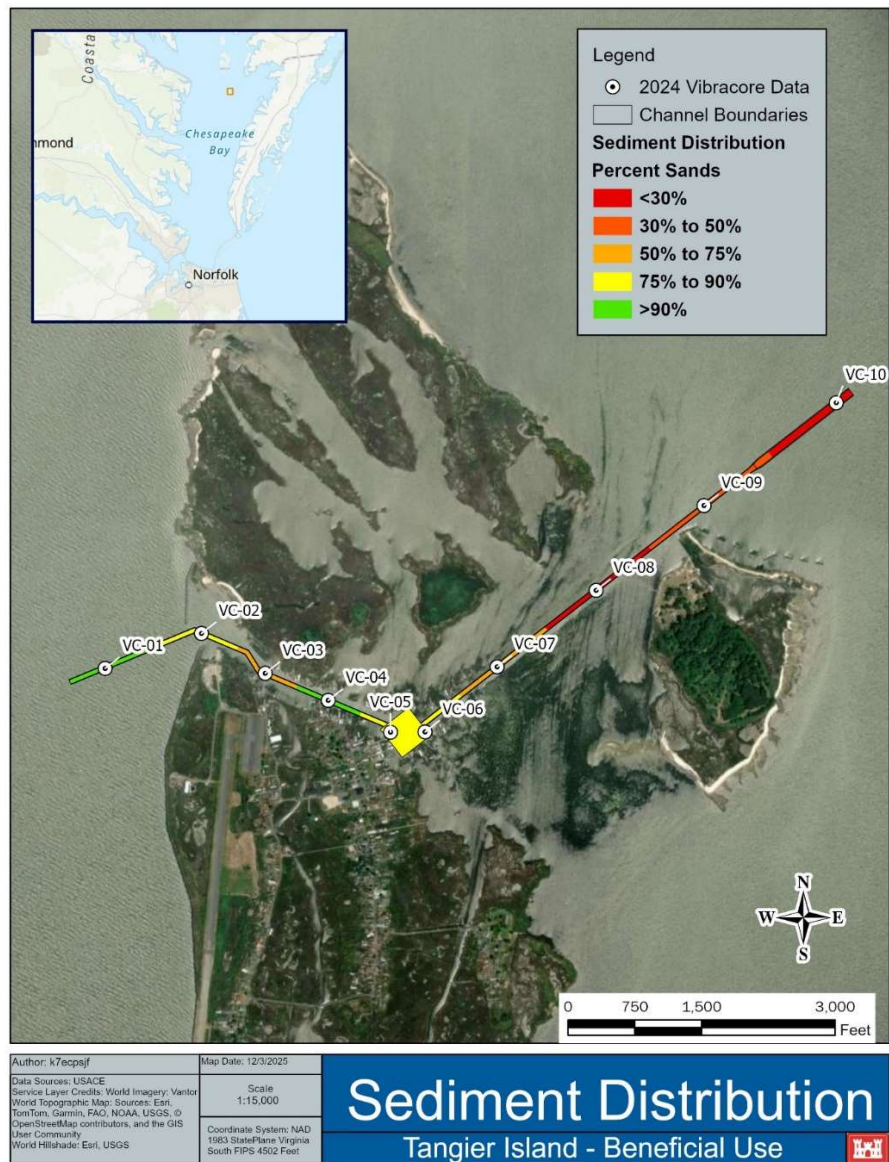


Figure 1. Sediment distribution throughout the Tangier Island navigation channel based on 2024 vibracore data.

Figure 2 presents a cross-section of vibracores collected throughout the channel. The visual classifications and laboratory results are not fully consistent, primarily due to data management issues and inconsistencies in the logger's application of sediment classification methods and laboratory sample selection. At this time, this represents the best available information for the project. Table 1 provides pertinent lab data for each sample, grain size curves and tables may be made upon request.

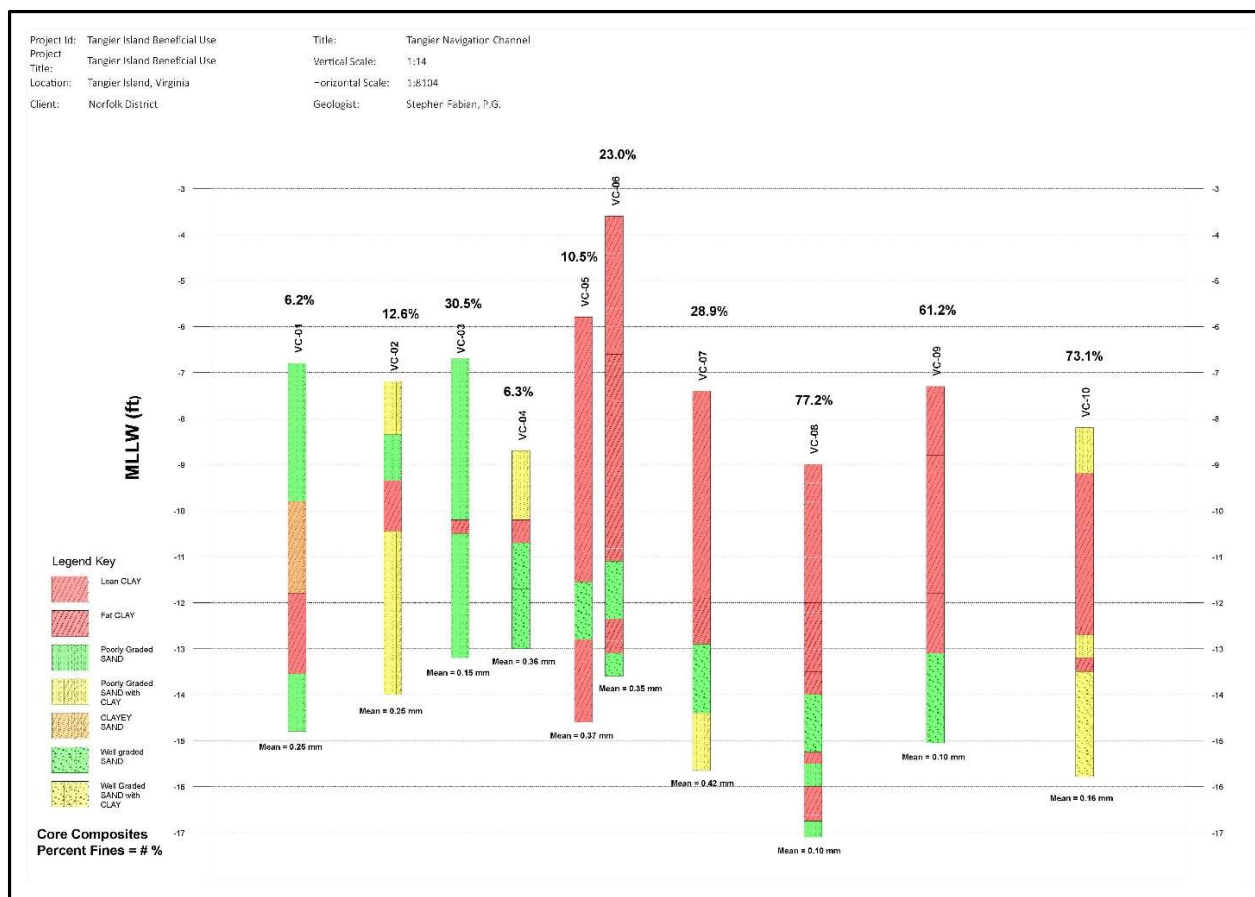


Figure 2: Vibracore cross-section from west to east across the navigation channel. Use Figure 1 to reference core locations.

Table 1. 2024 vibracore grain size data.

Core	D10 (mm)	D15 (mm)	D30 (mm)	D50 (mm)	D60 (mm)	D85 (mm)	Sorting (phi)	Mean (mm)	Passing (No. 200 Sieve)
VC-01	0.12	0.15	0.19	0.23	0.26	0.44	0.83	0.25	6.2%
VC-02	0.06	0.11	0.17	0.23	0.27	0.54	0.87	0.25	12.6%
VC-03	0.02	0.04	0.07	0.14	0.16	0.24	0.61	0.15	30.5%
VC-04	0.16	0.19	0.28	0.36	0.40	0.74	0.87	0.36	6.3%
VC-05	0.07	0.17	0.27	0.37	0.42	0.77	0.87	0.37	10.5%
VC-06	0.03	0.05	0.17	0.31	0.36	0.70	0.94	0.35	23.0%
VC-07	0.03	0.04	0.08	0.25	0.33	0.72	1.27	0.42	28.9%
VC-08	0.01	0.01	0.03	0.05	0.06	0.09	0.55	0.10	77.2%
VC-09	0.01	0.02	0.04	0.06	0.07	0.12	0.53	0.10	61.2%
VC-10	0.01	0.02	0.03	0.05	0.06	0.14	0.83	0.16	73.1%

Project Background – Coastal Summary

The west side of Tangier Island experiences some of the highest erosion rates within the lower Chesapeake Bay. Erosion is largely driven by storm events, especially nor'easters and tropical storms. The north-south orientation of the shoreline, combined with a large effective fetch averaging approximately 20 miles and extending just over 60 miles to the southwest, exposes the shoreline to sustained wind-driven wave energy. Wind observations recorded at the Tangier Island Airport indicate a predominance of winds from the north-northwest (NNW) and the south-southwest (SSW), (see Figure 3), which maximize wave development toward the western shoreline.

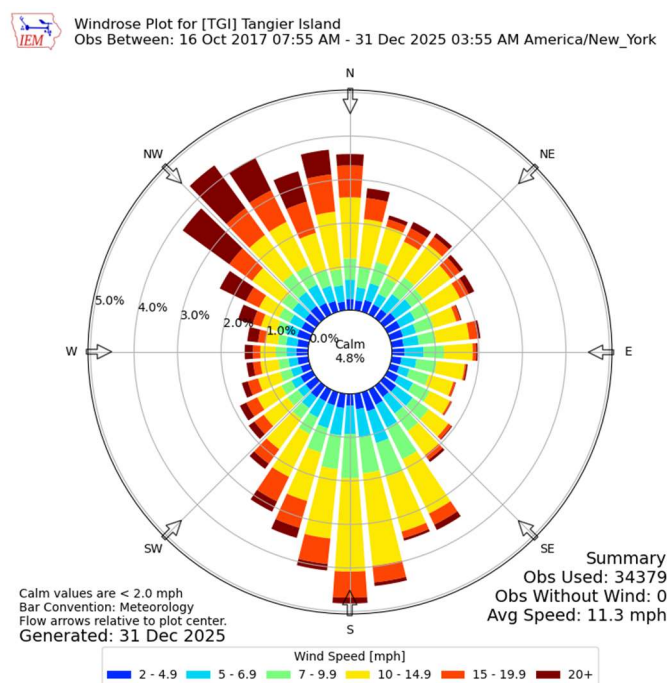


Figure 3 - Wind Observations at Tangier Island Airport

The tidal range of the region is relatively small, 1.5 feet between MLLW and MHHW. Figure 4 shows the tidal datum at the Lewisetta tide gauge. Water levels are strongly influenced by wind setup and storm surge commonly seen during tropical storms and nor'easters. During these events, the storm-driven waves combine with elevated water levels, increasing the potential for overtopping, toe scour, and structural instability. Table 2 below is the Annual Exceedance Probabilities for significant wave height and water levels, referenced from MLLW and adjusted to 2025 water levels.

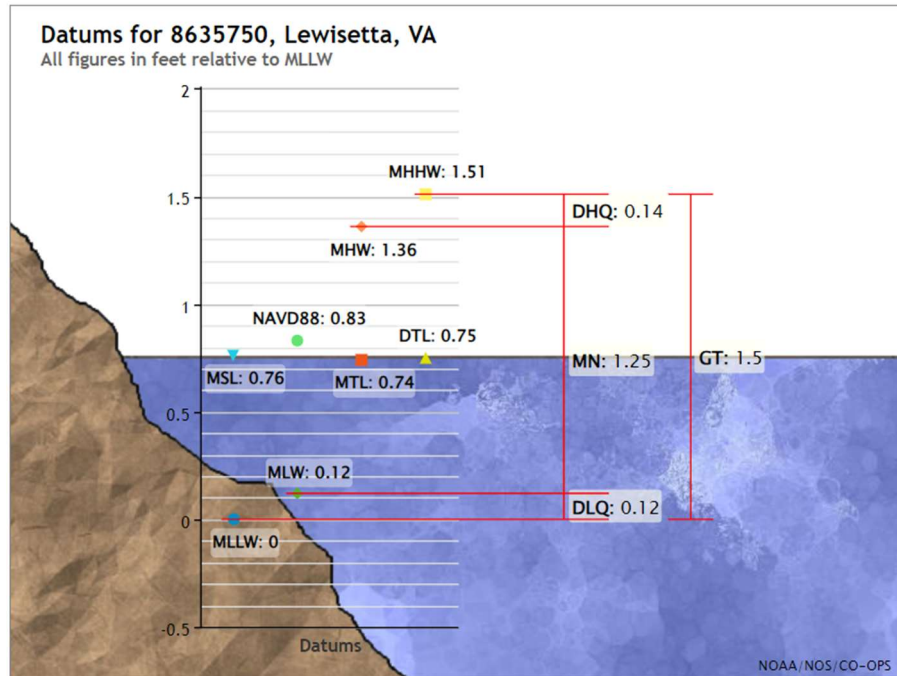


Figure 4 - Tide Datums for the Lewisetta, VA tide gauge, relative to MLLW

Table 2 - Annual Exceedance Values for Significant Wave Height and Water Levels, referenced to MLLW

Annual Exceedance Probabilities (AEP), referenced from Mean Lower Low Water (MLLW) 2025					
		Significant Wave Height (Hs)		Water Level	
AEP values	Return Period (years)	Expected Value AEP (m)	Expected Value AEP (ft)	Expected Value AEP (m)	Expected Value AEP (ft)
1	1	1.71	5.59	1.34	4.40
0.2	5	2.15	7.06	1.65	5.42
0.1	10	2.27	7.44	1.76	5.78
0.02	50	2.50	8.19	2.06	6.77
0.01	100	2.57	8.44	2.23	7.32

The nearshore bathymetry along the western side of Tangier is generally shallow, with gentle slopes and broad shoals near the island. This transitions into the deeper channel along the central portion of Chesapeake Bay. The shallow-water conditions, combined with long fetch exposure, result in relatively short-period, steeper waves that can enhance nearshore energy dissipation and contribute to shoreline erosion.